C:\hpc-23-feb-2026\.\30\eg202.cpp

```
#include<iostream>
#include<string>
#include<chrono>
#include<cstring>
#include<stdlib.h>
using namespace std;
#define isdigit(c) (c >= '0' && c <= '9')
int main()
{
  double a;
  int e;
  int c;
char buffer[1024];
char *s;
strcpy(buffer,"1231312.2334");
int x;
double value;
auto start=chrono::high_resolution_clock::now();
while(x<50000000)
{
 a=0.0;
 e=0;
 s=buffer;
 while ((c = *s++) != '\0' && isdigit(c)) {
   a = a*10.0 + (c - '0');
 }
 if (c == '.') {
   while ((c = *s++) != '\0' && isdigit(c)) {
     a = a*10.0 + (c - '0');
     e = e-1;
   }
 }
 if (c == 'e' || c == 'E') {
   int sign = 1;
   int i = 0;
   c = *s++;
   if (c == '+')
     c = *s++;
   else if (c == '-') {
     c = *s++;
     sign = -1;
   }
   while (isdigit(c)) {
     i = i*10 + (c - '0');
     c = *s++;
   }
   e += i*sign;
 }
```

```
  while (e > 0) {
    a *= 10.0;
    e--;
  }
  while (e < 0) {
    a *= 0.1;
    e++;
  }
x++;
}
auto end=chrono::high_resolution_clock::now();
auto d=chrono::duration_cast<chrono::milliseconds>(end-start);
cout<<d.count()<<" milliseconds"<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg199.cpp

```
#include<iostream>
#include<string>
#include<chrono>
using namespace std;
int main()
{
string s="1231312.2334";
int x;
double value;
auto start=chrono::high_resolution_clock::now();
while(x<50000000)
{
value=stod(s);
x++;
}
auto end=chrono::high_resolution_clock::now();
auto d=chrono::duration_cast<chrono::milliseconds>(end-start);
cout<<d.count()<<" milliseconds"<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg200.cpp

```
#include<iostream>
#include<string>
#include<chrono>
#include<stdlib.h>
using namespace std;
int main()
{
string s="1231312.2334";
int x;
double value;
auto start=chrono::high_resolution_clock::now();
while(x<50000000)
{
value=atof(s.c_str());
x++;
}
auto end=chrono::high_resolution_clock::now();
auto d=chrono::duration_cast<chrono::milliseconds>(end-start);
cout<<d.count()<<" milliseconds"<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg201.cpp

```
#include<iostream>
#include<string>
#include<chrono>
#include<stdlib.h>
using namespace std;
int main()
{
const char  *s="1231312.2334";
int x;
double value;
auto start=chrono::high_resolution_clock::now();
while(x<50000000)
{
value=atof(s);
x++;
}
auto end=chrono::high_resolution_clock::now();
auto d=chrono::duration_cast<chrono::milliseconds>(end-start);
cout<<d.count()<<" milliseconds"<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg203.cpp

```
#include<iostream>
#include<cstring>
using namespace std;
#define isdigit(c) (c >= '0' && c <= '9')
int main()
{
double a;
int e;
char c;
char buffer[1024];
strcpy(buffer,"1231231.544356,435345353.454565,435534.34534,3e20,345345.454\n");
char *s;
s=buffer;
a=0.0;
e=0;
while(1)
{
 while ((c = *s++) != ',' && c!='\n' && isdigit(c)) {
  a = a*10.0 + (c - '0');
 }
 if (c == '.') {
  while ((c = *s++) != ',' && c!='\n' && isdigit(c)) {
   a = a*10.0 + (c - '0');
   e = e-1;
  }
 }
 if (c == 'e' || c == 'E') {
  int sign = 1;
  int i = 0;
  c = *s++;
  if (c == '+')
   c = *s++;
  else if (c == '-') {
   c = *s++;
   sign = -1;
  }
  while (isdigit(c)) {
   i = i*10 + (c - '0');
   c = *s++;
  }
  e += i*sign;
 }
 while (e > 0) {
  a *= 10.0;
  e--;
 }
 while (e < 0) {
  a *= 0.1;
```

```
   e++;
  }
if(c==',' || c=='\n')
{
printf("%lf\n",a);
if(c=='\n') break;
a=0.0;
e=0;
}
}
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg204.cpp

```
#include<iostream>
#include<float.h>
using namespace std;
int main()
{
double a=DBL_MIN;
double b=DBL_MAX;
double c=-DBL_MAX;
// if you want to go for minimum possible value of double, use -DBL_MAX
cout<<a<<endl;
cout<<b<<endl;
cout<<c<<endl;
cout<<(a<c)<<endl;
cout<<(a==c)<<endl;
cout<<(c<a)<<endl;
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg205.cpp

```
#include<iostream>
using namespace std;
int main()
{
int x;
int num=0;
int upper_bound=31;
x=1;
while(x<=3)
{
cout<<num<<endl;
num++;
if(num==32) num=0;
if(num==0) x++;
}
return 0;
}
```

Thinking Machines Classroom Codes

C:\hpc-23-feb-2026\.\30\eg206.cpp

```
#include<iostream>
using namespace std;
int main()
{
int x;
int num=0;
int upper_bound=31;
x=1;
while(x<=3)
{
cout<<num<<endl;
num++;
num=num%32;  // costly
if(num==0) x++;
}
return 0;
}
```

C:\hpc-23-feb-2026\.\30\eg207.cpp

```
#include<iostream>
using namespace std;
int main()
{
int x;
int num=0;
int upper_bound=31;
x=1;
while(x<=3)
{
cout<<num<<endl;
num++;
num=num & 31;
if(num==0) x++;
}
return 0;
}
```

Thinking Machines Classroom Codes